

Chapter
Section Formula And Mid Point

1. The coordinates of the vertices of ΔABC are respectively $(-4, -2)$, $(6, 2)$ and $(4, 6)$. The centroid G of ΔABC is :

- (a) $(2, 2)$ (b) $(2, 3)$
(c) $(3, 3)$ (d) $(0, -1)$

Main 2023

2. The centroid of the triangle whose vertices are $(3, -7)$, $(-8, 6)$ and $(5, 10)$ is

- (a) $(0, 9)$ (b) $(0, 3)$
(c) $(1, 3)$ (d) $(3, 5)$

Comp 2023

3. If the point $C(-1, 2)$ divides internally the line segment joining $A(2, 5)$ and B in ratio $3 : 4$, find the coordinates of B .

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4. In what ratio does the point $P(2, -5)$ divide the line segment joining the points $A(-3, 5)$ and $B(4, -9)$?

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Solution

1.

Ans: (c) (3, 3)

Explanation:

The coordinates of the centroid $G(x, y)$ of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are given by:

$$G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Here,

$$x_1 = -4, x_2 = 6, x_3 = 4$$

$$y_1 = -2, y_2 = 2, y_3 = 6$$

$$G = \left(\frac{-4 + 6 + 4}{3}, \frac{-2 + 2 + 6}{3} \right)$$

$$G = \left(\frac{6}{3}, \frac{6}{3} \right)$$

$$G = (3, 3)$$

Thus, option (c) is the correct answer.

2.

Ans: (b) (0, 3)

Explanation:

The coordinates of the centroid $G(x, y)$ of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are given by:

$$G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Here,

$$x_1 = 3, x_2 = -8, x_3 = 5$$

$$y_1 = -7, y_2 = 6, y_3 = 10$$

$$G = \left(\frac{3 + (-8) + 5}{3}, \frac{-7 + 6 + 10}{3} \right)$$

$$G = \left(\frac{0}{3}, \frac{9}{3} \right)$$

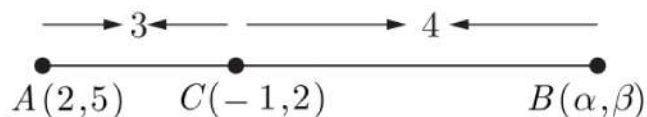
$$G = (0, 3)$$

Thus, option (b) is the correct answer.

3.

Ans:

Let the coordinates of point B be $B(\alpha, \beta)$.



Since point C divides the line segment AB internally in the ratio $3 : 4$,

By the **section formula**, the coordinates of C are:

$$C(x, y) = \left(\frac{3\alpha + 4 \times 2}{3 + 4}, \frac{3\beta + 4 \times 5}{3 + 4} \right)$$

$$C(x, y) = \left(\frac{3\alpha + 8}{7}, \frac{3\beta + 20}{7} \right)$$

But the coordinates of C are $(-1, 2)$.

Equating the corresponding coordinates:

$$\frac{3\alpha + 8}{7} = -1 \quad \text{and} \quad \frac{3\beta + 20}{7} = 2$$

$$3\alpha + 8 = -7 \quad \text{and} \quad 3\beta + 20 = 14$$

$$3\alpha = -15 \quad \text{and} \quad 3\beta = -6$$

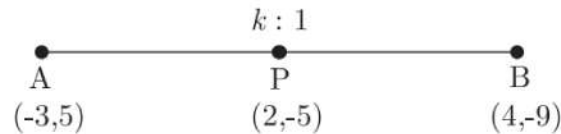
$$\alpha = -5 \quad \text{and} \quad \beta = -2$$

Hence, the coordinates of point B are

$$\boxed{(-5, -2)}$$

4.

Ans:



Let the point $P(2, -5)$ divide the line segment joining $A(-3, 5)$ and $B(4, -9)$ internally in the ratio $k : 1$.

By the **section formula**, the coordinates of P are given by:

$$P(x, y) = \left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1} \right)$$

Here,

$$A(x_1, y_1) = (-3, 5), \quad B(x_2, y_2) = (4, -9)$$

So,

$$P(2, -5) = \left(\frac{4k + (-3)}{k + 1}, \frac{-9k + 5}{k + 1} \right)$$

Equating the corresponding coordinates:

$$\frac{4k - 3}{k + 1} = 2 \quad \text{and} \quad \frac{-9k + 5}{k + 1} = -5$$

From the first equation:

$$4k - 3 = 2(k + 1)$$

$$4k - 3 = 2k + 2$$

$$2k = 5$$

$$k = \frac{5}{2}$$

From the second equation:

$$-9k + 5 = -5(k + 1)$$

$$-9k + 5 = -5k - 5$$

$$-4k = -10$$

$$k = \frac{5}{2}$$

Thus, the required ratio is

$$\boxed{5 : 2} \text{ (internally)}$$